
Mutual Self-Information and Mutual Information

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Review of Previous Lectures

1. Quantifying the amount of information carried by a specific message, by defining the self-information.
2. Quantifying the average amount of information generated by a discrete information source by defining the entropy.
3. Using source coding techniques to translate messages generated by the original information source and to reduce the redundancy of the original information source.

How to quantify the information through the channel?

Or we say, how much information is transmitted through the channel?

Main Content

1 Single-Symbol Discrete Channel

2 Mutual Self-Information

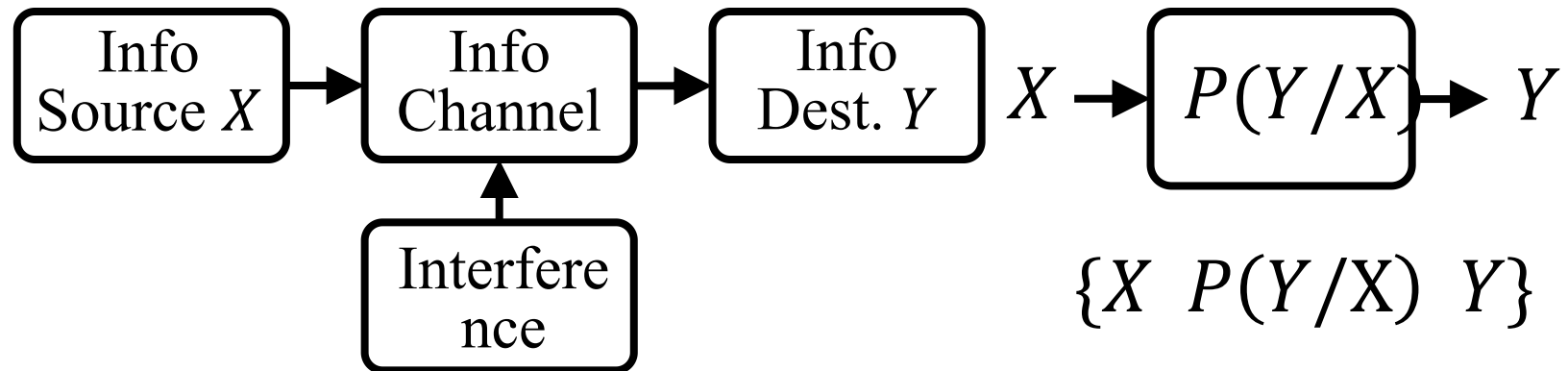
3 Mutual Information

4 Relationship among entropies





Discrete Channel Model



❑ How much information can be received by the destination if a message is transmitted through the information channel?

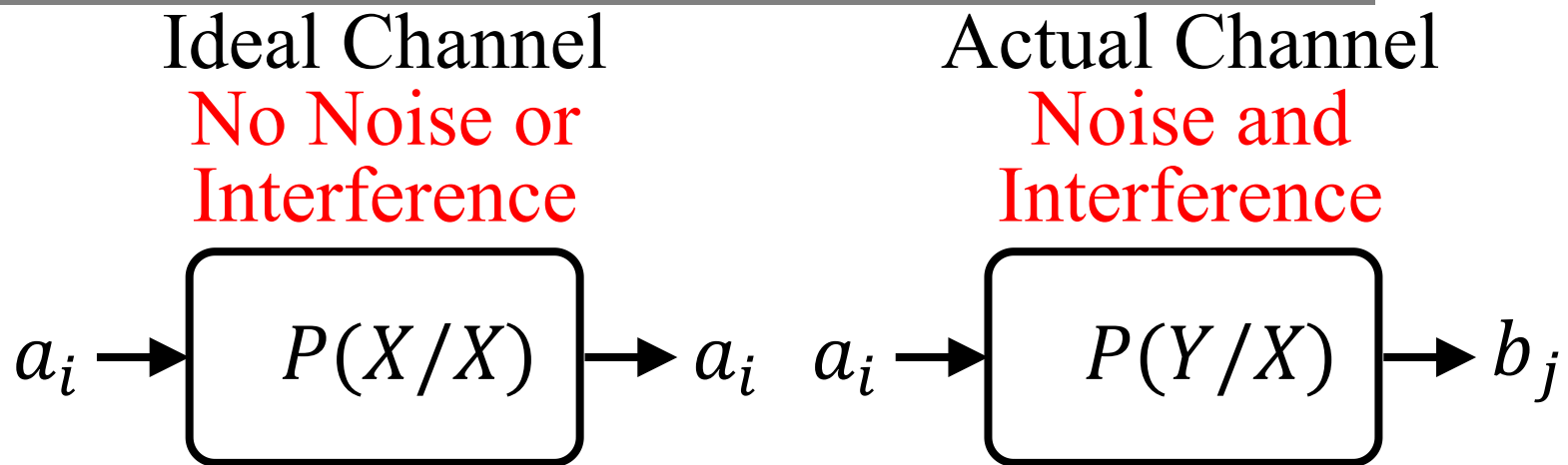
❑ The mathematical model of the information source X :

$$\begin{pmatrix} X \\ P(X) \end{pmatrix} = \begin{pmatrix} a_1, \dots a_i, \dots a_n \\ p(a_1), \dots p(a_i), \dots p(a_n) \end{pmatrix}; 0 \leq p(a_i) \leq 1, \sum_{i=1}^n p(a_i) = 1$$

❑ The mathematical model of the information destination Y :

$$\begin{pmatrix} Y \\ P(X) \end{pmatrix} = \begin{pmatrix} b_1, \dots b_j, \dots b_m \\ p(b_1), \dots p(b_j), \dots p(b_m) \end{pmatrix}; 0 \leq p(b_i) \leq 1, \sum_{j=1}^m p(b_j) = 1$$

Mutual Self-Info -- Definition



- ❑ The message/symbol received by the info destination is a random result chosen from a finite space.
- ❑ No noise or interference exists in the ideal channel. The message a_i can be losslessly received by the destination. The info. transmitted by the channel is that sent by the source.
- ❑ The transmission of a_i is distorted by the noise and interference channel. The message actually received by the destination is b_j . **How can we quantify the information transmitted by the channel?**

Mutual Self-Info -- Definition

Bayesian Inference

- ❑ After receiving the distorted message b_j (**evidence**), the **posterior probability** of the info destination inferring the original message a_i (**parameter**) is $p(a_i|b_j)$.
- ❑ The probability $p(a_i)$ of the info source sending the message a_i is the **prior probability**.
- ❑ Before transmitting the message a_i , the **likelihood** of the info source inferring the message b_j received by the info dest is $p(b_j|a_i)$.
- ❑ The **mutual self-information** between b_j received by the info dest and a_i sent by the info source is defined as the division between the **posterior probability** $p(a_i|b_j)$ and the **prior probability** $p(a_i)$:

$$I(a_i; b_j) = \log_2 \frac{p(a_i|b_j)}{p(a_i)}$$

Example

- We model the final of World Cup 2018, France versus Croatia in 90 minutes of the regular time as a discrete information source:

$$\begin{pmatrix} X \\ P(X) \end{pmatrix} = \begin{pmatrix} a_1(\text{Win}), a_2(\text{Draw}), a_3(\text{Lose}) \\ 50\%, \quad 29\%, \quad 21\% \end{pmatrix}$$

When we receive a message $b_1 = \text{“France is unbeaten in 90 mins”}$, please calculate the mutual self-information between the received message b_1 and all the possible original messages sent.

- The weather in February is modelled by the following info source:

$$\begin{pmatrix} X \\ P(X) \end{pmatrix} = \begin{pmatrix} a_1(\text{sunny}), a_2(\text{cloudy}), a_3(\text{rainy}), a_4(\text{snowy}) \\ 1/2, \quad 1/4, \quad 1/8, \quad 1/8 \end{pmatrix}$$

When we receive a message $b_1 = \text{“Today is not sunny”}$, please calculate the mutual self-information between the received message b_1 and the original messages a_2 sent.

Physical Meaning of Mutual Self-Info

- Observe from the **channel output**, the mutual self-info between the message b_j received and that a_i sent is expressed as

$$\begin{aligned} I(a_i; b_j) &= \log_2 \frac{p(a_i/b_j)}{p(a_i)} = -\log_2 p(a_i) + \log_2 p(a_i/b_j) \\ &= I(a_i) - I(a_i/b_j) \end{aligned}$$

Self-info quantifies the uncertainty of a message being sent. The difference represents the uncertainty reduction, which is also the information about a_i acquired from b_j and the information transmitted in the channel.

- Observe from the **channel input**, the mutual self-info between the message a_i sent and that b_j received is expressed as

$$I(b_j; a_i) = \log_2 \frac{p(b_j/a_i)}{p(b_j)} = I(b_j) - I(b_j/a_i)$$

Observer reduces some uncertainty about b_j by having the prior knowledge of a_i . The uncertainty reduction is the information transmitted in the channel.

Physical Meaning of Mutual Self-Info

- Observe from the **entire comm system, before any transmission**, we do not know that the source and the dest are correlated. The joint probability of a_i and b_j appearing simultaneously at the channel input and the output, respectively, is $p'(a_i b_j) = p(a_i)p(b_j)$, the corresponding total self-info is

$$I'(a_i b_j) = -\log_2 p'(a_i b_j) = I(a_i) + I(b_j)$$

- After the transmission, the source and the dest are connected by the channel. The joint probability of a_i and b_j appearing simultaneously at the channel input and the output, respectively, is $p(a_i b_j) = p(a_i)p(b_j|a_i)$. The self-info **after the transmission** is then expressed as

$$I(a_i b_j) = -\log_2 p(a_i b_j)$$

- The **uncertainty reduction** is the mutual self-information transmitted in the channel:

$$I(a_i; b_j) = I'(a_i b_j) - I(a_i b_j) = I(a_i) + I(b_j) - I(a_i b_j) = \log_2 \frac{p(a_i b_j)}{p(a_i)p(b_j)}$$

Mutual Self-Info – Properties

- ❑ The mutual self-info is **symmetric**, which is expressed as

$$I(a_i; b_j) = I(b_j; a_i),$$

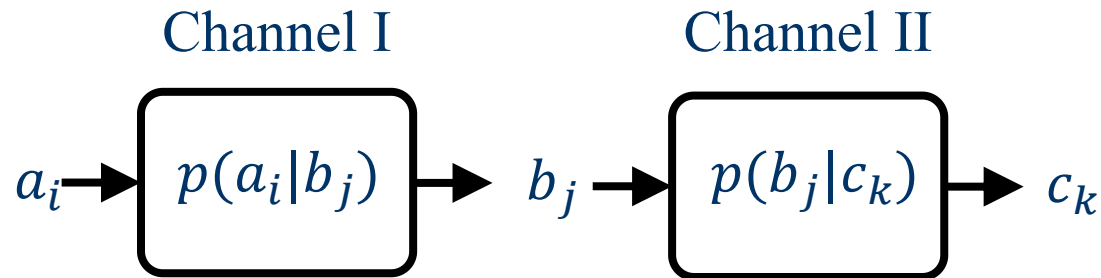
which indicates that the information $I(a_i; b_j)$ about a_i contained in b_j is equal to the information $I(b_j; a_i)$ about b_j contained in a_i .

- ❑ When the info source X is **independent** of the info dest Y , the mutual self-info $I(a_i; b_j)$ is **zero**, which indicates that a_i and b_j are uncorrelated to each other. We cannot acquire any info about a_i from b_j and vice versa.

- ❑ The mutual self-info could be positive or negative, which is expressed as

$$I(a_i; b_j) = \log_2 \frac{p(a_i/b_j)}{p(a_i)} = \begin{cases} + & p(a_i/b_j) > p(a_i) & \text{Info Transmission reduces the uncertainty.} \\ 0 & p(a_i/b_j) = p(a_i) & \text{No Info Transmission} \\ - & p(a_i/b_j) < p(a_i) & \text{Interference increases the uncertainty} \end{cases}$$

Conditional Mutual Self-Info



□ Given c_k , the **conditional mutual self-info** between a_i and b_j is defined as

$$I(a_i; b_j | c_k) = \log_2 \frac{p(a_i | b_j c_k)}{p(a_i | c_k)}$$

Conditional Mutual Self-Info

- The mutual self-info between the message a_i and the joint message $b_j c_k$ is

$$I(a_i; b_j c_k) = \log_2 \frac{p(a_i | b_j c_k)}{p(a_i)}$$

- Given c_k , the **conditional mutual self-info** between a_i and b_j is defined as

$$I(a_i; b_j | c_k) = \log_2 \frac{p(a_i | b_j c_k)}{p(a_i | c_k)}$$

- According to the definition of the mutual self-info, we may obtain

$$I(a_i; b_j c_k) = I(a_i; c_k) + I(a_i; b_j | c_k),$$

The info about a_i
contained in the
joint message $b_j c_k$

The info about
 a_i contained in
the message c_k

Given c_k , the info
about a_i contained
in the message b_j

Mutual Self-Info and Mutual Info

- ❑ Mutual self-info quantifies the info transmitted between **a certain message pair**. Different message pairs have different mutual self-info. Mutual self-info cannot statistically quantify the information transmitted between **a pair of info source and destination**.
- ❑ The joint probability of the source sending a_i and the destination receiving b_j is $p(a_i b_j)$, the corresponding mutual self-info is $I(a_i; b_j)$.
- ❑ The mutual info is defined as the average of the mutual self-info $I(a_i; b_j)$ on the space of all the possible joint messages, which is defined as

$$\begin{aligned} I(X; Y) &= \sum_{i=1}^n \sum_{j=1}^m p(a_i b_j) I(a_i; b_j) \\ &= \sum_{i=1}^n \sum_{j=1}^m p(a_i b_j) \log_2 \frac{p(a_i | b_j)}{p(a_i)}. \end{aligned}$$

This is observed from the **channel output (info dest) Y** .

Physical Meaning of Mutual Info

- Observe from the information destination Y , the mutual info is

$$I(X; Y) = \sum_{i=1}^n \sum_{j=1}^m p(a_i b_j) \log_2 \frac{p(a_i | b_j)}{p(a_i)}$$

$$= H(X) - H(X|Y) \leftarrow \text{The information loss due to the imperfect transmission of the channel, loss entropy}$$

Prior Uncertainty of the info source X

Given the info dest Y , the posterior uncertainty of X , channel equivocation

Mutual info represents that given the posterior knowledge of the info dest Y , the uncertainty reduction of the info source X , which is also the information about X contained in Y .

Physical Meaning of Mutual Info

□ Observe from the information source X , the mutual info is

$$I(Y; X) = \sum_{i=1}^n \sum_{j=1}^m p(a_i b_j) \log_2 \frac{p(b_j | a_i)}{p(b_j)}$$

$$= H(Y) - H(Y|X)$$

Prior Uncertainty of
the info destination Y

Given the info source X ,
the posterior uncertainty
of the info destination Y .

←----- The uncertainty
induced by the
channel noise,
noise entropy.

Mutual info represents that given the prior knowledge of the info source X , the uncertainty reduction of the info dest Y , which is also the information about Y contained in X .

Physical Meaning of Mutual Info

- Observe from the entire communication system, the mutual info is

$$I(X; Y) = \sum_{i=1}^n \sum_{j=1}^m p(a_i b_j) \log_2 \frac{p(a_i b_j)}{p(a_i) p(b_j)}$$

$$= \boxed{H(X) + H(Y)} - H(XY)$$

Before info transmission,
the total uncertainty of
the comm system.

The channel results in the
correlation between X and
 Y . The total uncertainty of
the comm system after the
info transmission.

Mutual info represents that the uncertainty reduction of the entire comm system after the info transmitted from the source X to the destination Y , which is info about X (or Y) contained in Y (or X).

Example

- The messages sent of the info source

$$\begin{pmatrix} X \\ P(X) \end{pmatrix} = \begin{pmatrix} a_1 & a_2 \\ 0.6 & 0.4 \end{pmatrix}$$

are transmitted by a interference channel. The received alphabet of the info dest is $Y = \{b_1, b_2\}$. The **channel transition matrix** is

$$\begin{bmatrix} 5/6 & 1/6 \\ 1/4 & 3/4 \end{bmatrix}.$$

Please calculate:

- Self-info of the messages a_1 and a_2 sent by the info source X ;
- Self-info of a_i , when b_j is received;
- Entropies of the info source X and the info dest Y ;
- Channel equivocation $H(X/Y)$ and the noise entropy $H(Y/X)$;
- Mutual information $I(X; Y)$.

Mutual Info -- Properties

- The mutual info is **symmetric**, which is expressed as

$$I(X; Y) = I(Y; X).$$

This can be proved by exploiting the symmetry of the mutual self-info $I(a_i; b_j) = I(b_j; a_i)$ and the definition of the mutual info. For X and Y connected by the channel, the info of X contained in Y is equal to the info of Y contained in X , which are both the info transmitted in the channel.

- The mutual info is **non-negative**, which is expressed as

$$I(X; Y) \geq 0$$

This can be proved by exploiting the definition of the mutual info (observe from the info dest) and the classic inequality $\ln x \leq x - 1$. When messages are transmitted in the channel, there must be some info delivered to the dest. In the worst case, zero info is received by the dest. Secure communication is an exception, in which the mutual info could be negative.

Mutual Info -- Properties

- The extremum of the mutual info is expressed as

$$I(X; Y) \leq H(X)$$

$$I(Y; X) \leq H(Y)$$

which are proved by exploiting the non-negativity of the conditional entropy.
When the info channel connecting X and Y is perfect:

$$p(a_i/b_j) = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

We have $I(X; Y) = H(X)$. We may know all the information about X , given the posterior knowledge of Y , which represents that all the info of the source X is transmitted in the channel without any loss. When the source X is independent of the dest Y , namely $p(a_i/b_j) = p(a_i)$, the mutual info is $I(X; Y) = 0$. In this case, we cannot obtain any information of one random event from the other, which represents the outage of the channel.

Mutual Info' s Functional Expression

- ❑ Mutual info $I(X; Y)$ is a function of the info source's probability distribution $\{p(a_i), i = 1, 2, \dots, n\}$ and the channel transition probability distribution $\{p(b_j/a_i), i = 1, 2, \dots, n; j = 1, 2, \dots, m\}$, namely

$$I(X; Y) = f[\{p(a_i)\}, \{p(b_j/a_i)\}]$$

- ❑ Given a specific channel, the mutual info $I(X; Y)$ is a function of the info source's probability distribution $\{p(a_i), i = 1, 2, \dots, n\}$:

$$I(X; Y) = f'[\{p(a_i)\}]$$

- ❑ Given a specific source, the mutual info $I(X; Y)$ is a function of the channel's transition probability distribution $\{p(b_j/a_i), i = 1, 2, \dots, n; j = 1, 2, \dots, m\}$:

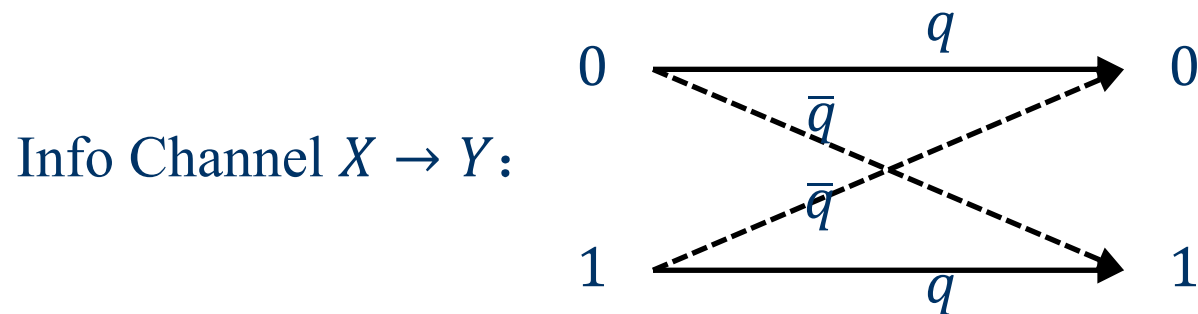
$$I(X; Y) = f''[\{p(b_j/a_i)\}]$$

- ❑ The functions $f'[\{p(a_i)\}]$ and $f''[\{p(b_j/a_i)\}]$ have different characteristics.

Example

□ We have the following binary info source X and the info channel $X \rightarrow Y$:

Info Source X :
$$\begin{pmatrix} X \\ P(X) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ p & \bar{p} = 1 - p \end{pmatrix}$$



Where $\bar{q} = 1 - q$. Please calculate the mutual info.

Mutual Info -- Convexity

- Mutual Info $I(X; Y)$ is a **concave function** of the probability distribution $\{p(a_i), i = 1, 2, \dots, n\}$ of the info source.

Proof: We have two probability distributions for an info source $\{a_1, a_2, \dots, a_n\}$, namely $P_1 = \{p_1(a_i), i = 1, 2, \dots, n\}$ and $P_2 = \{p_2(a_i), i = 1, 2, \dots, n\}$. For $0 < \alpha < 1$, we have to prove the following inequality:

$$f[\alpha P_1 + (1 - \alpha)P_2] \geq \alpha f[P_1] + (1 - \alpha)f[P_2]$$

- Mutual Info $I(X; Y)$ is a **convex function** of the transition probabilities $\{p(b_j/a_i), i = 1, 2, \dots, n; j = 1, 2, \dots, m\}$ of the info channel.

Proof: We have two channel transition probability distributions for an info channel $X \rightarrow Y$, namely $Q_1 = \{p_1(b_j/a_i), i = 1, 2, \dots, n; j = 1, 2, \dots, m\}$ and $Q_2 = \{p_2(b_j/a_i), i = 1, 2, \dots, n; j = 1, 2, \dots, m\}$. For $0 < \alpha < 1$, we have to prove the following inequality (**Use Log-Sum Inequality**):

$$f[\alpha Q_1 + (1 - \alpha)Q_2] \leq \alpha f[Q_1] + (1 - \alpha)f[Q_2]$$

Convexity



Concave Upward

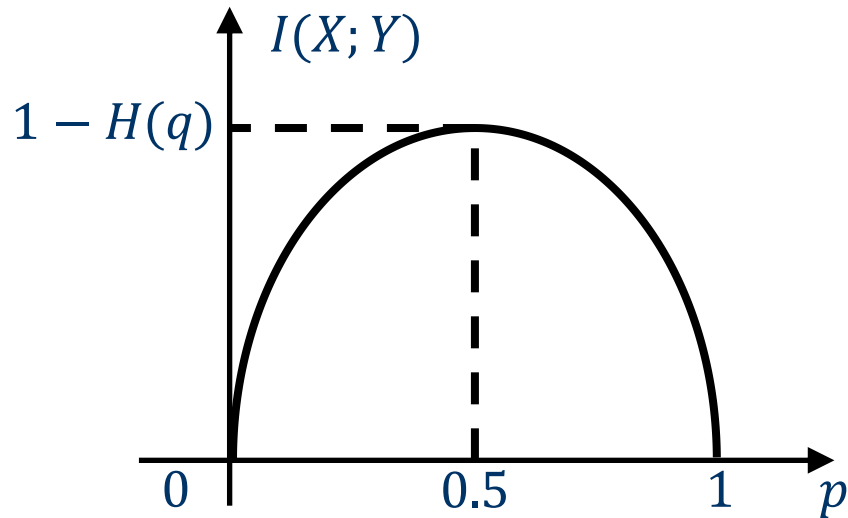
Also called "Convex"
or "Convex Downward"



Concave Downward

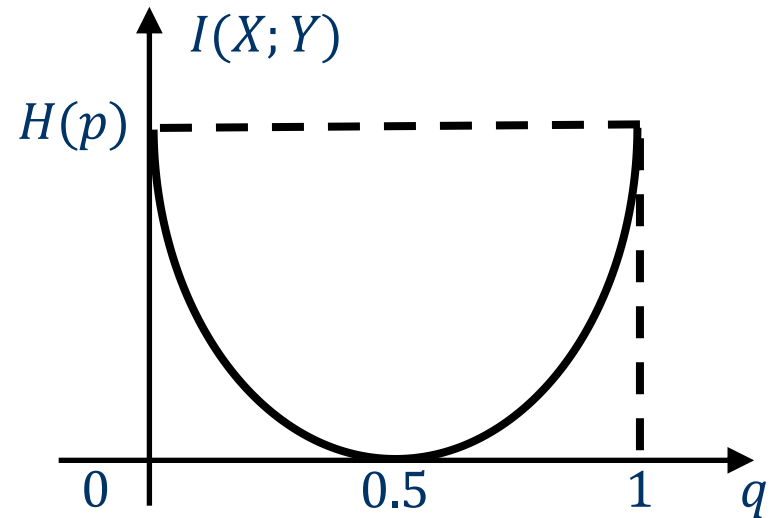
Also called "Concave"
or "Convex Upward"

Mutual Info -- Convexity



Mutual info against the probability distribution of the info source, when a specific channel is given.

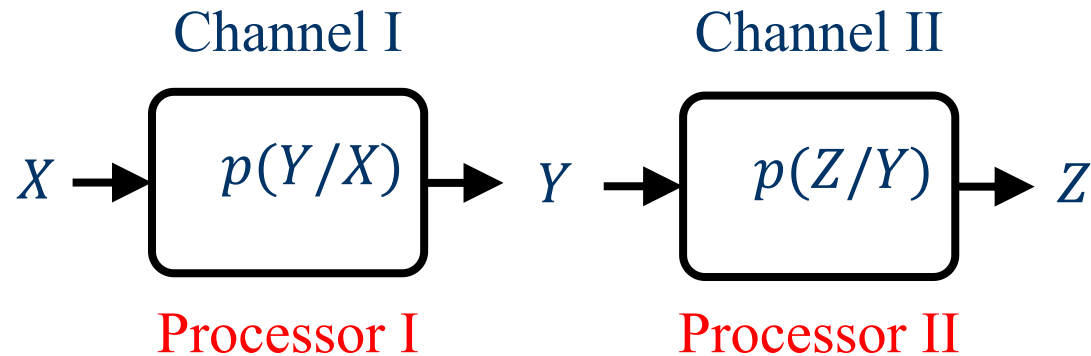
Fundamental of
channel capacity



Mutual info against the transition probability of the info channel, when a specific source is given.

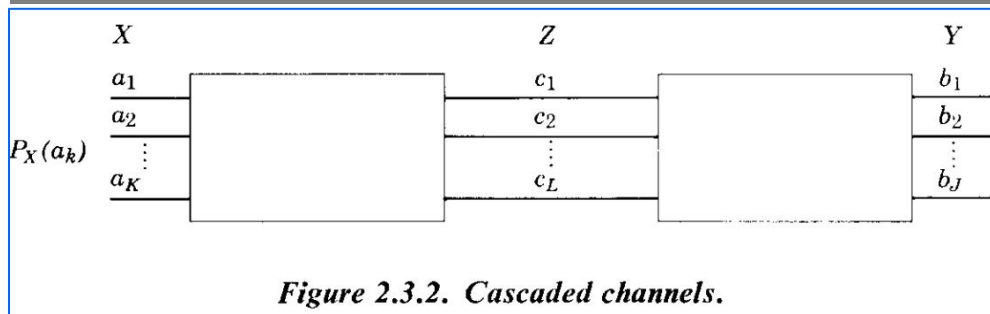
Fundamental of rate
distortion theory

Data-Processing Inequality



- ❑ **Data-processing theorem:** when the info is processed by a sequence of processors, the mutual info between the input and the output tends to be reduced.
- ❑ **Data-processing inequality:** We have a pair of concatenated processors (channels), where X and Z are independent of each other when Y is given (X , Y and Z constitute a Markov chain), then we have $I(X; Z) \leq I(X; Y)$.
- ❑ **Data-processing inequality shows that** no clever manipulation of the data can improve the inferences that can be made from the data. Data processing only convert data into a more useful form but it cannot create new information.

Can the Cascaded Channel increase the average mutual information?



$$I(X;YZ) = I(X;Y) + I(X;Z | Y)$$

$$= I(X;Z) + I(X;Y | Z)$$

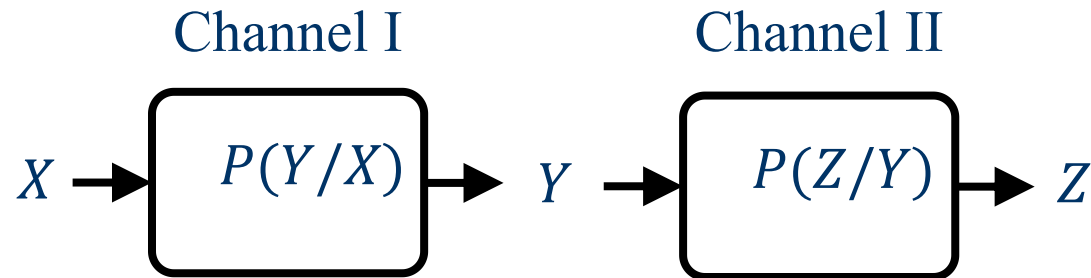
$$\left. \begin{array}{l} P(x, y | z) = P(x | z)P(y | z) \\ H(X | Z) \geq H(X | ZY) \end{array} \right\} \Rightarrow I(X;Y | Z) = 0$$

$$\Rightarrow I(X;Z) = I(X;Y) + I(X;Z | Y)$$

$$\Rightarrow \left\{ \begin{array}{l} I(X;Z) \geq I(X;Y) \\ I(Y;Z) \geq I(X;Y) \end{array} \right. \Rightarrow \begin{array}{l} H(X) - H(X | Z) \geq H(X) - H(X | Y) \\ H(X | Z) \leq H(X | Y) \end{array}$$

The derivation yields that the intuitively satisfying result that this uncertainty or equivocation can never decrease as we go further from the input on a sequence of cascaded channels.

Example



- We have a info source $X: \begin{pmatrix} X \\ p(X) \end{pmatrix} = \begin{Bmatrix} a_1 & a_2 \\ 0.5 & 0.5 \end{Bmatrix}$, the info channel I having an reverse transition probabilities

$$P(X/Y) = \begin{Bmatrix} p(a_1/b_1) & p(a_1/b_2) & p(a_2/b_1) & p(a_2/b_2) \\ 0.8 & 0.2 & 0.2 & 0.8 \end{Bmatrix}$$

the info channel II having an reverse transition probabilities

$$P(Y/Z) = \begin{Bmatrix} p(b_1/c_1) & p(b_1/c_2) & p(b_2/c_1) & p(b_2/c_2) \\ 0.9 & 0.1 & 0.1 & 0.9 \end{Bmatrix}$$

Please calculate the mutual info $I(X; Y)$, $I(Y; Z)$, $I(X; Z)$.

Multi-Measurements on Info Dest

- ❑ One measurement of the mutual info on info dest:

$$I(X; Y_1) = H(X) - H(X/Y_1)$$

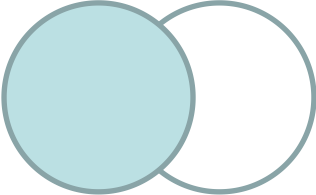
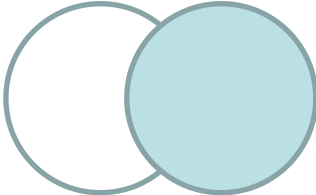
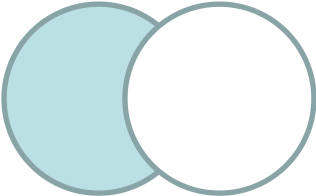
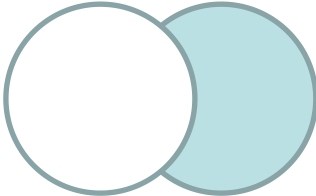
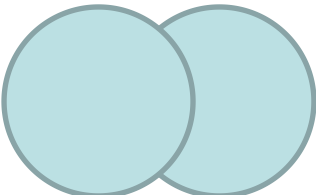
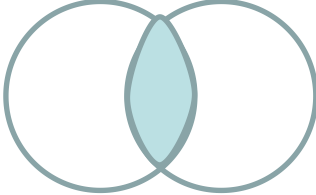
- ❑ Two measurement of the mutual info on info dest:

$$I(X; Y_1 Y_2) = H(X) - H(X/Y_1 Y_2)$$

We can prove: $H(X/Y_1 Y_2) \leq H(X/Y_1)$, namely $I(X; Y_1 Y_2) \geq I(X; Y_1)$.

- ❑ Having more measurements on info destination, the conditional entropy of the info source X is reduced until tending to zero.
- ❑ No matter how we measure the info dest Y , the info obtained is not higher than the source entropy $H(X)$.
- ❑ Once we obtain Y , no matter how we process Y , the information obtained is not higher than $I(X; Y)$ (data-processing inequality).

Entropies and Mutual Info

$H(X)$		$H(Y)$	
$H(X/Y)$		$H(Y/X)$	
$H(XY)$ $H(YX)$		$I(X; Y)$ $I(Y; X)$	

https://www.bilibili.com/video/BV1PV4y1u7Ub/?spm_id_from=333.337.search-card.all.click&vd_source=edb0b420c3d23a94c634fea505172a7d